

ISyE 6525 High Dimensional Data Analytics - HW4

Module 4: Optimization Methods I

Fall 2026

Submission. Submit *one PDF report per question*, and for every question that requires code, *one code file for that question*. You may use MATLAB, R or Python. Total: **100 points**.

Question 1.

(20 points)

Convex sets and convex functions.

(a) **(6 pts)** Prove that the following sets are convex.

- i. $\{\mathbf{x} : \|\mathbf{x}\|_2 \leq 1\}$, where $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$.
- ii. $\{\mathbf{x} : A\mathbf{x} = b\}$, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ are fixed.

(b) **(8 pts)** Prove that the following functions are convex.

- i. Exponential: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^{ax}$, for any $a \in \mathbb{R}$.
- ii. Negative logarithm: $f : \{x : x > 0\} \rightarrow \mathbb{R}$, $f(x) = -\log x$.
- iii. Non-negative weighted sum: $f(\mathbf{x}) = \sum_{i=1}^k \omega_i f_i(\mathbf{x})$, where $f_1, \dots, f_k$ are convex and $\omega_1, \dots, \omega_k \geq 0$.

(c) **(6 pts)** Determine whether $f(x_1, x_2) = \frac{x_1^2}{x_2}$ with $\text{dom } f = \{(x_1, x_2) : x_2 > 0\}$ is convex. Justify your answer.

Question 2.

(45 points)

Maximum likelihood estimation. Consider the likelihood function

$$L(k, c; y_1, \dots, y_n) = \prod_{i=1}^n \frac{k c y_i^{c-1}}{(1 + y_i^c)^{k+1}},$$

where $k > 0$ and $c > 0$ are unknown parameters and the observations $y_i > 0$ are in `Question2.csv` ($n = 2000$ observations, one per row).

- (a) **(5 pts)** Write down the log-likelihood function $\ell(k, c)$.
- (b) **(3 pts)** Write down the corresponding maximum likelihood formulation.
- (c) **(12 pts)** Derive the gradient $\nabla \ell(k, c)$ and the Hessian $\nabla^2 \ell(k, c)$. *Show your work.*
- (d) **(20 pts)** Implement each of the four methods below yourself; do **not** call `fminunc`, `optim`, `scipy.optimize` or any equivalent solver. Estimate $\hat{k}_{\text{MLE}}$ and $\hat{c}_{\text{MLE}}$ with each, starting from $(k_0, c_0) = (1, 1)$. Plot the value of the log-likelihood and of each parameter against the iteration number, and report the final estimates.

1. Gradient descent (exact line search)
2. Accelerated gradient descent (exact line search)
3. Stochastic gradient descent (one randomly drawn observation per update, fixed step size, as in the slides)
4. Newton's method (exact line search)

Note: y_i^c overflows for moderately large c . Rewrite the expressions using $\log(1 + y^c) = \log(1 + e^{c \log y})$ and $\frac{y^c}{1 + y^c} = \frac{1}{1 + e^{-c \log y}}$ before coding them.

- (e) **(5 pts)** Compare the four methods in terms of the number of iterations needed and the cost of one iteration. Which one would you use here, and would your answer change if the model had 500 parameters instead of 2?

Question 3.

(35 points)

Leak estimation in transmission systems. Leakage of transmission fluid in a power-train casting can cause overheating and permanent damage, so every part is leak-tested. The leak flow is modelled as a function of both the testing time and the part temperature,

$$F_{ij} = (\beta_0 + \beta_1 T_i) \left(1 - \exp(-2\beta_2 t_{ij})\right) + \epsilon_{ij}, \quad \epsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2),$$

where F_{ij} is the flow measured at time t_{ij} on a part held at temperature T_i . `Question3.csv` contains 80 measurements with columns `time`, `temperature` and `flow`. The goal is to estimate $\beta = (\beta_0, \beta_1, \beta_2)^\top$ by minimizing

$$L(\beta) = \sum_i \sum_j \left\| F_{ij} - (\beta_0 + \beta_1 T_i) (1 - \exp(-2\beta_2 t_{ij})) \right\|_2^2.$$

- (a) **(5 pts)** The loss can be written as $L(\beta) = g(\beta)^\top g(\beta)$. Define $g(\cdot)$ explicitly and state its dimension.
- (b) **(10 pts)** Derive the Jacobian matrix $J_g(\beta)$ of g .
- (c) **(5 pts)** Express the gradient ∇L and the Hessian $\nabla^2 L$ in terms of J_g and g , and state precisely which term the Gauss-Newton method drops and when that approximation is good.
- (d) **(15 pts)** Implement the Gauss-Newton algorithm yourself, with an adaptive step size, and estimate β starting from $\beta^{(0)} = (1, 0.1, 1)^\top$. Report the estimates, the number of iterations, and the root mean square error overall and for each temperature. Plot the loss against the iteration number and the fitted curves against the data.